

PHYSICAL CONDITIONING FOR CLIMBING

Physical conditioning, or fitness as it's commonly referred to, is essential for improving performance and preventing injury in any sport. Hörst (2006/2004) suggests the following breakdown of time spent on each type of training: *Beginner*: 70 per cent climbing, 25 per cent general physical conditioning, 5 per cent strength-building exercises specific to climbing.

Intermediate: 60 per cent climbing, 35 per cent strength-building exercises specific to climbing, 5 per cent general physical conditioning.

Elite/high performance: 35 per cent climbing, 60 per cent strength-building exercises specific to climbing, 5 per cent general physical conditioning.

Spending too much time and energy on physical conditioning is a common mistake, especially among novice climbers. What's more, it is often overly and unnecessarily strenuous, which can be a direct route to overloading key muscles and, in all likelihood, an unwelcome injury. Remember: 'don't train more: train better'.

One of the key messages of this book is that physical conditioning, especially in relation to strength, shouldn't cause a level of fatigue that limits or prevents us from climbing (which is what we actually want to be doing).

STRENGTH TRAINING TO MAXIMISE PERFORMANCE. METHODS AND TECHNIQUES FOR UPPER BODY, CORE AND LOWER BODY EXERCISES

At this point, I think we can agree that maximum strength training (low reps, high loads, max speed) is essential to improve climbing performance. However, some exercises are more important than others (they more directly transfer to climbing) and they shouldn't all be done at the same intensity (upper body pulling strength, which we can train by doing pull-ups, will always have a greater impact on performance than lower body pushing strength, which we can train by doing deadlifts). Although we've already touched on this, the gold standard for strength training is the VBT method, but not everyone has access to a VBT device. As such, the percentage of intra-set loss of speed can be used to calculate the intensity of training expressed as an effort level (EL) – number of reps(maximum number of reps possible), for example 5(10) – and a percentage of your 1RM (adapted from González-Badillo *et al.*, 2017):

Beginner (fewer than two seasons of training):

Loss of speed: 10–20 per cent

EL: 8(20) to 6(12)

Percentage of your 1RM: 55–70 per cent

Intermediate (more than two seasons of training):

Loss of speed: 10–25 per cent

EL: 8(18) to 4–5(7–8)

Percentage of your 1RM: 57–80 per cent

Advanced:

Loss of speed: 15–25 per cent

EL: 8(16) to 1–3(2–4)

Percentage of your 1RM: 60–90 per cent



BENCH PRESS

1



BENCH PRESS

2



BENCH PRESS

3

Remember that since the aim is to build *useful* strength – strength that is of real and not just aesthetic benefit – our total volume of training or the number of sets that we do should be based on our ability to maintain speed and effort level as opposed to sticking to any fixed number or volume.

Upper body exercises:

Bench press: the upper body pushing exercise par excellence. Although the bench press is usually thought of as a pectoral exercise, as with any exercise and especially multi-joint exercises, it involves a much wider range of muscles. The exercise consists of lying flat on a bench and pushing a bar above your chest.

The antagonist or target muscles include the pectoralis major, anterior deltoid and triceps, while the pectoralis minor, rhomboids, trapezius, latissimus dorsi and even the biceps act in synergy with these muscles to stabilise the scapula and the shoulder joint (Rippetoe, 2011). The involvement of each muscle varies a lot depending on technique and the weight being lifted. For example, with weights of around 40 per cent 1RM, the split is 40 per cent pectoral muscles, 40 per cent shoulder muscles and 20 per cent triceps; whereas at 90 to 95 per cent 1RM, it's 29.5 per cent pectoral muscles, 40.5 per cent shoulder muscles and 30 per cent triceps (Marchante, 2015).



PUSH-UP PLUS

1



PUSH-UP PLUS

2



PUSH-UP PLUS

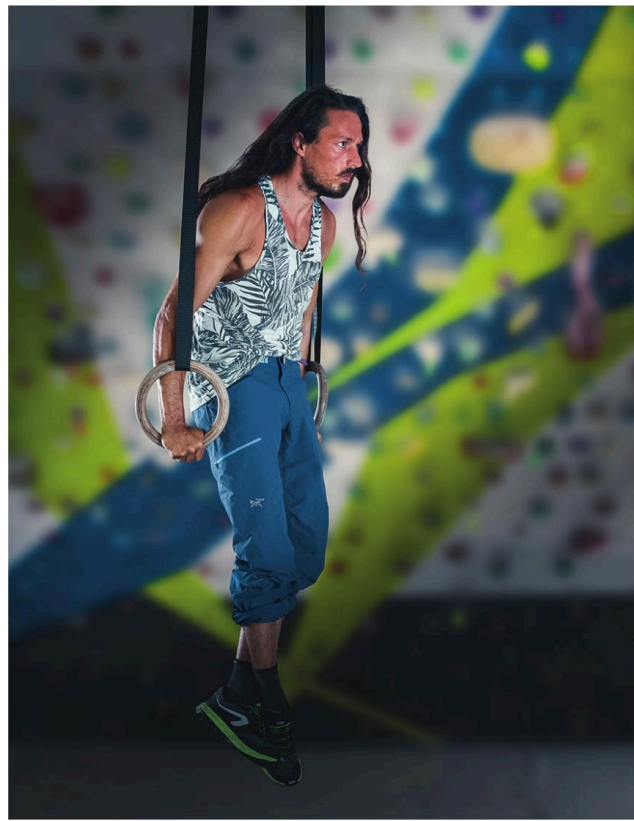
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To do a bench press, lie with your back on a weight bench and with your eyes level with the bar. Your feet should be firmly grounded on the floor at a comfortable distance apart, roughly the same as for a squat, and your shins should be perpendicular to the floor. Your back should be straight, meaning there should be a natural curve at the base of the spine. Don't try to lift your legs and/or support the lower back as flattening this natural curve will weaken the lower back and make it harder for the intervertebral discs to absorb the load. The upper back shouldn't be totally flat either, as the scapulae need to be retracted to protect the shoulders (PowerExplosive, 2015). As for grip position, the narrower you go, the more you target the triceps over the pectoral muscles. The ideal distance is shoulder width plus a palm's width on each side: this will keep the forearms perpendicular to the floor throughout the lift. Once you're in position, take hold of the bar, stabilise and bring it down to lightly rest on the chest. Then lift the bar in a 'J-shaped' motion, pushing up and slightly towards the face (Marchante, 2015). The main difference between using a bar and free weights is stability. Free weights have a lot more movement than a bar held in position by both hands, meaning you'll need to use lighter weights as a lot of your strength will go into stabilising the lift with free weights.



DIPS PLUS

1



DIPS PLUS

2

Push-up plus: push-ups are an excellent alternative if you don't have the equipment to do a bench press. Place your hands slightly wider than shoulder-width apart, with fingers pointing forwards. Your shoulders should be externally rotated and down away from your ears. It's really important to fully engage your core throughout this exercise to prevent over-extending the lower back and endangering the spine. Begin in a plank position and, with your body in a straight line, lower down until your chest touches the floor, without losing body tension or resting your weight on the floor. To add the 'plus', push back up until your arms are straight and continue pushing up to protract the scapula (the opposite of retraction), rounding out through the shoulders and the upper back. This small additional movement really activates the serratus anterior, which helps stabilise the scapula.

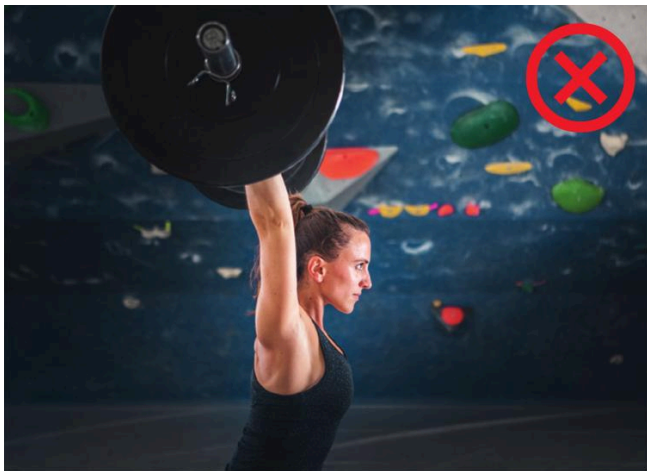
Dips plus: dips are incredibly transferable to climbing, especially for mantels. With this exercise, the further forwards your chest, the more you'll activate the pectoral muscles, while a more upright position will target the triceps. Your hands should be far apart enough for the wrist and elbow to be perfectly aligned, from a front-on view, for the whole movement. Lower until your shoulders dip just below elbow height. If you've got the mobility, you can go lower but it should never be painful (Marchante, 2015). It's best to do dips on a stable surface so you can apply as much force as possible. You can also do them on rings (as in the photos) but this will significantly increase the demand on the shoulder-stabilising muscles. As with push-ups, you can add the 'plus' to target the serratus anterior.



MILITARY PRESS 1



MILITARY PRESS 2



MILITARY PRESS
DON'T STRAIN YOUR HEAD FORWARDS.



MILITARY PRESS
AVOID BEHIND-THE-NECK PRESS TO PROTECT YOUR SHOULDERS.

Military press: also known as overhead press. In this exercise, you start with the bar in line with your collar bones and lift it above your head. As the military press is performed standing up, it requires full body tension, creating a chain of muscles that transfers force from your feet all the way up to your hands (Marchante, 2015; Rippetoe, 2011). As such, it's a good way to work on transferring force from the lower to the upper body, while actively engaging the core and the arms. To do a military press, lightly rest the bar on your collar bones with your elbows perpendicular to the floor. Push upwards until your arms are straight and the bar is above your head. Make sure you don't strain your head forwards; if you can't keep the bar in line with your body due to poor shoulder mobility, you'll need to address this first (try flossing before this exercise if your range of motion is limited). Also be aware of pushing your hips forwards or curving your lower back: your body should be in a straight, stable position with knees slightly bent and a neutral pelvis and spine. To complete the lift, carefully lower the bar back to your collar bones. As for the behind-the-neck press (starting and/or finishing with the bar behind your head), unless you're looking for a nasty shoulder impingement (Michener *et al.*, 2003), I'd recommend avoiding this exercise as the slight extra activation of the anterior deltoid in no way outweighs the really high risk of injury.

Pull-ups: an exercise that all climbers do, but not all climbers do right. You should always begin by retracting the scapulae to avoid compressing the subacromial space. Then think about pulling the bar to your chest, not your body to the bar, working through the full range of movement for this exercise. This means hanging with straight arms, pulling until your chin is above the bar and lowering all the way back down to straight arms, maintaining scapular retraction throughout. Starting or finishing with bent arms or not getting your chin above the bar will create sticking points, where you'll fail to develop any strength (Marchante, 2015). As a result, you'll notice a real weakness in these positions when climbing. In terms of grip width, a study by Prinold and Bull (2016) found that very wide pull-ups limit scapular mobility, reducing stability and compressing the subacromial space. In terms of grip position, a pronated grip (palms facing away from you – what's commonly known as a pull-up) gives less mechanical advantage but greater recruitment of the brachioradialis, trapezius and latissimus dorsi (Dickie *et al.*, 2017). What's more, a pronated grip has the greatest transfer to climbing. Interestingly, you get better activation of

the biceps, brachioradialis and pectoralis major with a neutral grip (palms facing each other) than with a supinated grip (palms facing back towards you – commonly known as a chin-up). And the only advantage of a supinated grip over a neutral grip is slightly greater activation of the latissimus dorsi (Dickie *et al.*, 2017).

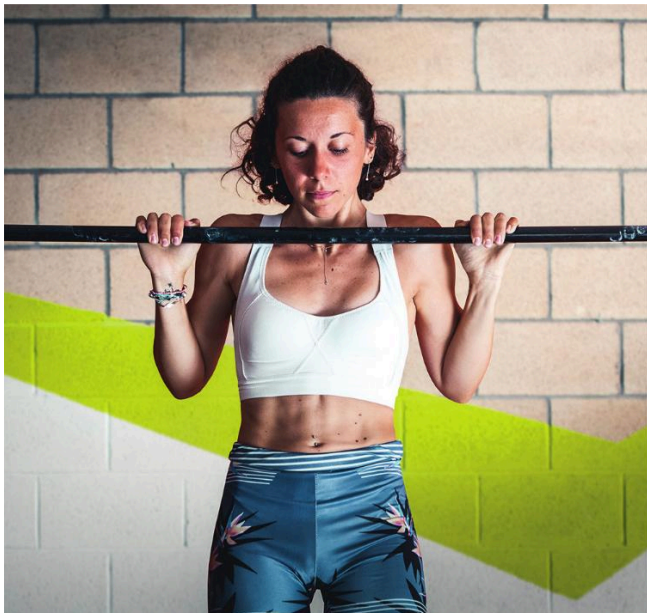
Barbell row: barbell rows, with the hands at varying widths apart and mainly using a pronated grip position, are excellent for strengthening the latissimus dorsi, trapezius, teres major, infraspinatus and posterior deltoid muscles (Marchante, 2015). As such, while this lift activates the latissimus dorsi and the internal rotator muscles, it also activates many of the muscles used to stabilise the scapulae. To begin, stand with your back flat and parallel to the floor. Knees should be slightly bent and arms should hang perpendicular to the floor. Grip the bar and pull it up towards you to touch the lower chest/upper abdomen, before lowering back down in a controlled manner (see [images on the next page](#)). As you lift the bar, retract your scapulae, keep your shoulders down away from the ears and your elbows tucked in. The wider apart your hands, the greater the demand for scapular stabilisation. If you switch to a supinated or neutral grip, this will increase activation of the biceps.



PULL-UPS



MAINTAIN SCAPULAR RETRACTION



PULL-UPS

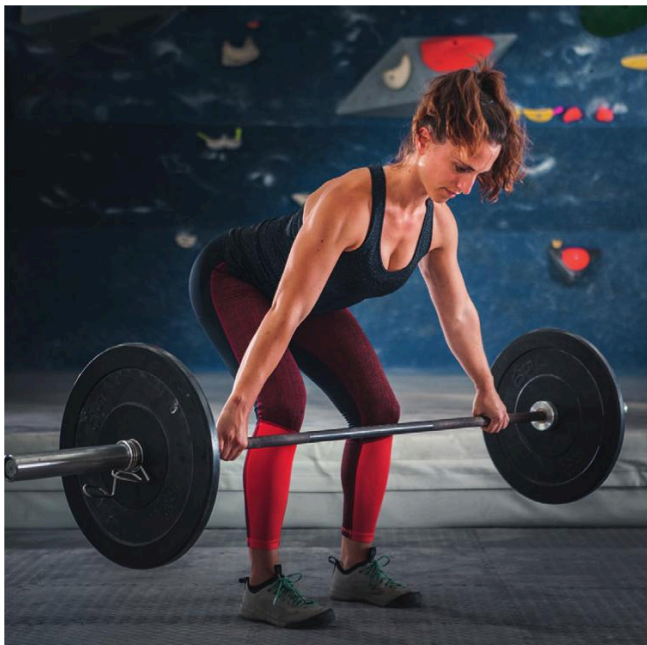
Face pull: this exercise really targets the posterior deltoids and the muscles involved in external shoulder rotation. Begin by holding a resistance band or a pulley at between chest and head height. Pull the band towards your face, separating your hands as you pull back. Keep your elbows bent and arms parallel to the floor, a bit like a double-bicep bodybuilding pose (Marchante, 2015). Keep your shoulders down and try to pinch the scapulae together as much as possible at the top of the pull. This is a brilliant exercise for scapular stability and injury prevention.



BARBELL ROW 1



BARBELL ROW 2



BARBELL ROW WIDE GRIP 3



Core exercises:

Core training is a delicate subject. Firstly, what's advertised (mainly by the fitness industry) as core training is rarely the best way to activate the core: countless balancing acts on unstable surfaces, where the limiting factor is more likely to be the ankle, knee or hip; all sorts of acrobatics; and absolutely any type of TRX exercise, whether it activates the abdomen or not. In a systematic review of multi-joint exercises, both Rogan *et al.* (2014) and Martuscello *et al.* (2013) found that squats or deadlifts activated the core much more than traditional exercises like sit-ups, planks, and so on. Logically, if the main function of the core is to stabilise the spine and transfer force between the upper body and the lower body, core training needs to in some way challenge the stability of the lower back or force the transfer of energy from one end of the body to the other. For example, if you're doing squats at 90 per cent of your 1RM, say with 160 kilograms, and your core *isn't* fully engaged, you're probably going to put your back out! That said, in defence of single-joint exercises, they do allow you to feel and activate different parts of your core without the 'distractions' of multi-joint exercises. However, just remember that, according to research, multi-joint or compound movements give greater muscle activation and overall gains.



FACE PULL

1



FACE PULL

2

Isometric exercises will strengthen the stabilising function of the core.

Plank: in this exercise, the core acts like a bridge and the abdominal muscles are the keystone, working to stabilise the lower back. The load on the lower back is similar to when trying to keep your feet on when climbing on an overhang. To perform a plank correctly, on your hands or your forearms, you need to press your heels together (activating the external rotator muscles and the deep hip muscles), activate your quads and squeeze your glutes. The pelvis should be neutral, although if you lack strength and/or motor control, a posterior pelvic tilt (keeping the pelvis rotated back and down) will maximise activation of the rectus abdominis and help maintain a flat and neutral spine. It's also important to draw your navel in, activating the transverse abdominis, and to push down through your hands or forearms to separate the scapulae and activate the serratus anterior. Keep your head in line with your body and your neck in a neutral position (PowerExplosive, 2016). In addition, the spine should be long and neutral (see top image on the opposite page) to properly activate the multifidus muscle. You can reduce the intensity by placing your hands on a stable, raised surface (a block or step), your knees on the floor or your feet out wide for a more stable base (see central images).

To increase the intensity, you can remove points of contact (raising a hand or a foot, or the opposite hand and foot at the same time – bottom left image), place your hands or feet on an exercise ball or an unstable surface, or ask a friend to push you from the hips in different directions or to pull on a resistance band looped around your ankle (bottom right image).



PLANK



PLANK: LOW INTENSITY



PLANK: LOW INTENSITY



PLANK: HIGH INTENSITY



PLANK: HIGH INTENSITY



PALLOF PRESS



Pallof press: in this exercise, the oblique abdominal muscles stabilise the spine by preventing rotation. In climbing, this is similar to the anti-rotational force needed to stop a barn door swing. Your feet should be no more than hip-width apart, so with roughly the width (not the length!) of one foot between them. The closer together your feet, the harder the exercise. Keep your knees soft, with a slight bend, and the pelvis neutral, maintaining the natural curve of

the lower back. Squeeze the scapulae together and drop your shoulders, keeping your head properly aligned (not strained forwards) and the spine long and neutral. In this position, hold on to a resistance band at chest height. Ask a friend to stand to one side and pull the band tight (or you can anchor it to the wall), then extend your arms out in front of you. This creates a bigger lever and greater demand on the core. The aim of the exercise is to resist the rotational pull of the band and remain in the same forward-facing position (see images). To increase the intensity, you could make a smaller base (with feet close together), change the angle of the band or create intermittent tension.



ABDOMINAL CRUNCH

If these exercises are done correctly, in addition to stabilising the lower back, you will have correctly aligned the spine and moved the shoulder girdle into the best possible position. This effectively teaches the body that in the event of similar external forces, it is the *core* which should absorb these forces while also transferring them to other parts of the body. These positions don't need to be held for very long. In fact, holding a plank for a whole minute has very little transfer to climbing or any other sport: can you think of a move where you're under full body tension, without moving, for a whole minute? Logically, long-hold isometric exercises don't make a lot of sense. Instead, it's better to do them for no more than 12 to 15 seconds and to increase the intensity of the exercise instead (adding unstable surfaces, changes in tension at different angles, and so on), returning once again to the idea that greater maximum strength equals greater submaximal strength (PowerExplosive, 2016).

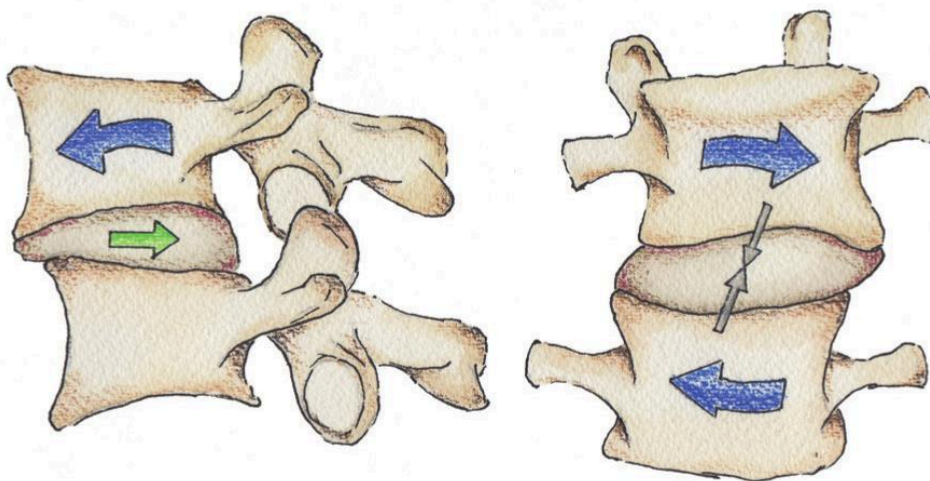
In turn, dynamic core exercises strengthen both the stabilising function of the core and the transfer of force. However, some of the most popular dynamic exercises come with a significant risk of injury. Let's look at some examples:

Abdominal crunch: an incredibly popular exercise, advertised and performed all over the world. While it may not be the most harmful, it's definitely one of the worst options for improving core strength. Simply put, it has very little transfer not only to climbing or any other sport, but to any aspect of daily life. Can you think of a sport that mimics this movement? The adverse effects of the abdominal crunch are well documented: increased pressure on the nucleus pulposus (the gel-like centre of the intervertebral discs) (Arshad *et al.*, 2016), which pushes the discs backwards and compresses the spinal canal (Davies, 2008); increased intra-abdominal pressure, which has an adverse effect on the pelvic floor; increased ligament stress and tension in the longus colli muscle, which alters the natural curve of the cervical spine (Davies, 2008); and increased pull on the psoas, which compresses and compromises the intervertebral discs in the lumbar spine (Grenier and McGill, 2007; Muscolino, 2014). These combined effects could easily result in a hernia, prolapse or urinary incontinence, so clearly abdominal crunches are not an option for strengthening the core. This goes for both the most concentrated version, where you only raise the scapulae off the floor to target the 'upper abs' (a training fallacy, see [chapter 4](#)) and the classic sit-up (which mainly works the iliopsoas, not the rectus abdominis), which we've probably all done in a PE class, strength test or gym assessment.



RUSSIAN TWIST

Russian twist: this trendy exercise is great ‘cross training’ and very ‘functional’ because it uses a kettlebell (note the irony here!). In reality, it alters the natural curve of the spine, puts pressure on the psoas, compromises the lower back, creates scapular instability and is usually performed with very little control. What’s more, over 50 years ago, Kapandji had already linked the combination of spinal flexion and spinal torsion to shearing of the intervertebral discs in his renowned book *The Physiology of the Joints*. In short, this is a textbook hernia-inducing exercise.



GRAPHICS: JAVIER FERNÁNDEZ DE CARA

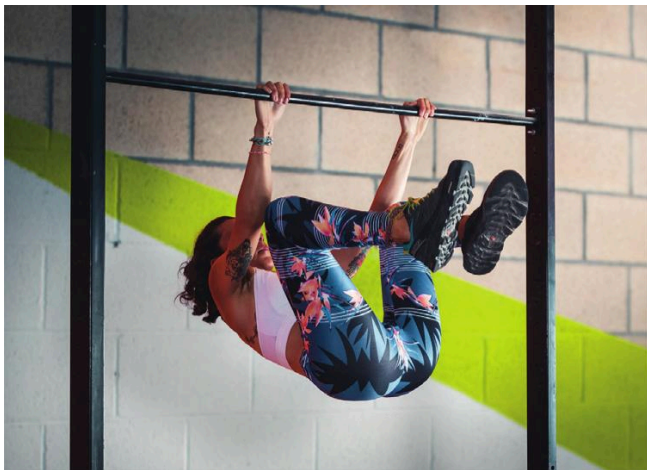
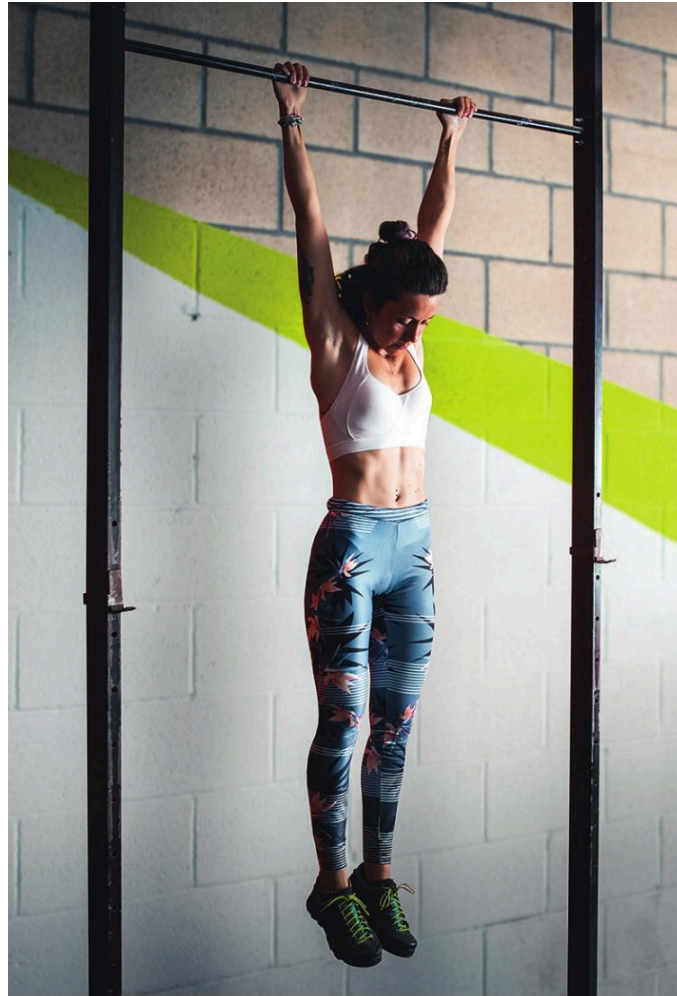


LOWER BACK EXTENSION

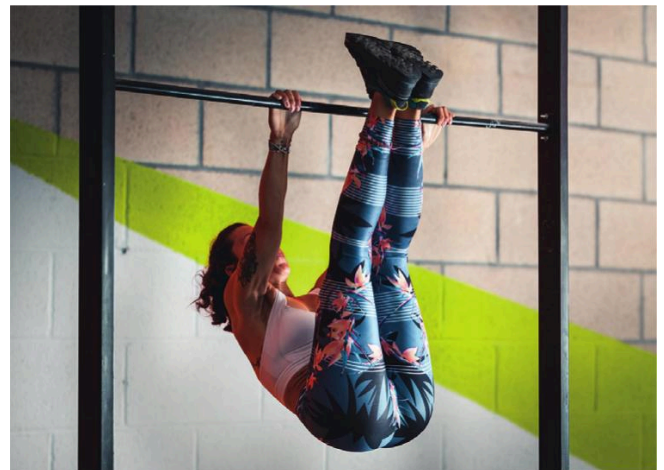
Lower back extension: whether on the floor or using a back-extension bench, I wouldn't advise this exercise either, because it compresses the intervertebral discs, puts pressure on the facet joints between the vertebrae, causes pelvic instability and can lead to ligament inflammation. The glute bridge is a much better alternative as it activates and synchronises the gluteus maximus and the lower back muscles, making it much more transferable to sport and general health.

There are, however, some exercises that, despite their potential for injury, could be beneficial or even necessary for climbers, such as toes to bar, or knees to elbows.

These exercises are risky because hanging from a bar could compress the subacromial space, compromising the supraspinatus tendon. What's more, the main muscle worked in these exercises is the psoas, and with poor motor control, you'll end up overloading the lower back. That said, with a few modifications (maintaining good scapular retraction, engaging the transverse abdominal muscle, engaging the latissimus dorsi as you lift your legs, and lowering with control to prevent hyper-extending the lumbar spine), these exercises have great transfer to climbing.



KNEES TO ELBOWS



TOES TO BAR

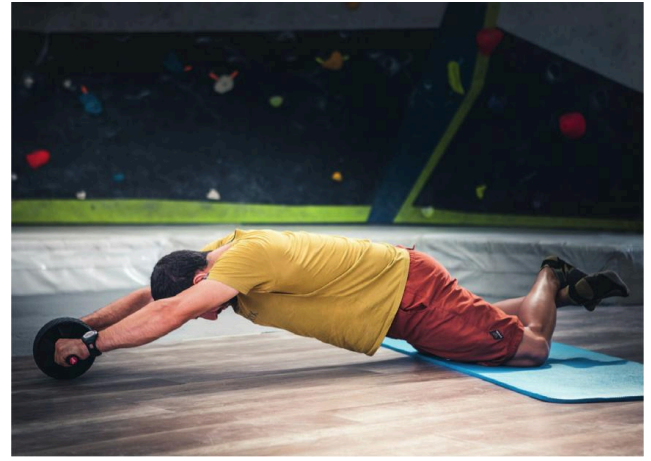
Fortunately, there are a number of safe and efficient dynamic exercises which can build core strength.

Ab roll-outs (or TRX roll-outs): the best exercise for activating the rectus abdominis (Calatayud *et al.*, 2017). If you're new to core training, you could start with a kneeling roll-out and a posterior pelvic tilt (keeping the pelvis rotated back and down) to encourage activation and prevent overloading the lower back. The next step would be a kneeling roll-out with the pelvis in a neutral position. If you have a very strong core, you could do a standing roll-out, without using your knees. With any variation, it's important to keep the hips straight (a straight line from shoulders to the point of contact with the floor) and to pivot from the shoulders. Finally, rolling out really far won't give you greater activation if you can't stabilise your lower back. As a guide, if your lower back hurts when you do this exercise, you're rolling too far.



AB ROLL-OUT

1



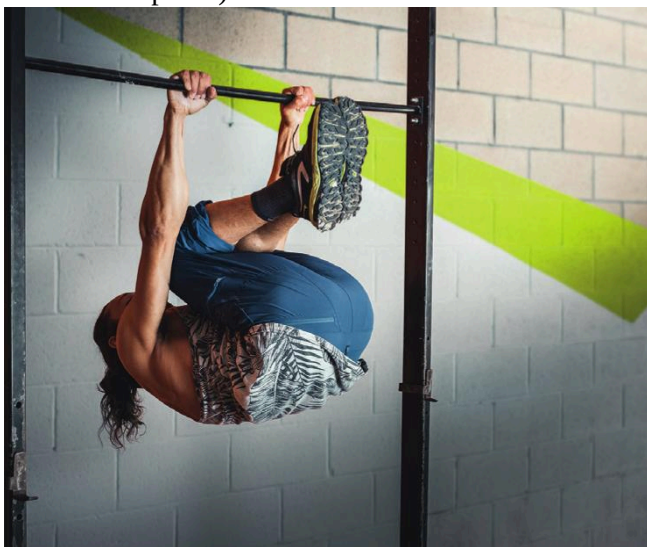
AB ROLL-OUT

2

Front lever: more advanced climbers may be able to do a front lever, which is a very demanding core exercise. In terms of progression, there are lots of different options, so I'll stick with the same method we used for training pull-ups: the eccentric method. (Like with pull-ups, if you aren't able to do a front lever, it's because it's beyond your max strength.) I'll also explain a bit about the physics of levers.

In a front lever, the fulcrum or pivot point is your shoulders; the effort arm is mainly the rectus abdominis (as well as the transverse abdominis, latissimus dorsi, multifidus and the pectoral muscles ... it's a full body tension exercise); and the load arm is your own body. The longer and heavier the load arm, the harder it is on the abdominal muscles. The first step is to hold a front tuck: both knees tucked to your chest with your back parallel to the floor – this brings the hips and the legs as close as possible to the fulcrum. From here, gradually lengthen the load arm by extending one leg. Once you can fully extend one leg with the other tucked against your chest, the next step is to extend both legs in a straddle position (which reduces the total length of the load arm). The final step is to gradually bring your legs together until you can hold a perfect front lever.

Farmer's walk: walking along while holding a heavy weight (sack, free weights, kettlebell) on one or both sides of the body. This exercise works absolutely every part of your core at all different angles as you walk along (you could even do this when you're out shopping by carrying a heavy bag in each hand and moving in different directions and at different speeds).



FRONT LEVER PROGRESSION

1



FRONT LEVER PROGRESSION

2



FRONT LEVER PROGRESSION

3



FRONT LEVER



FARMER'S WALK



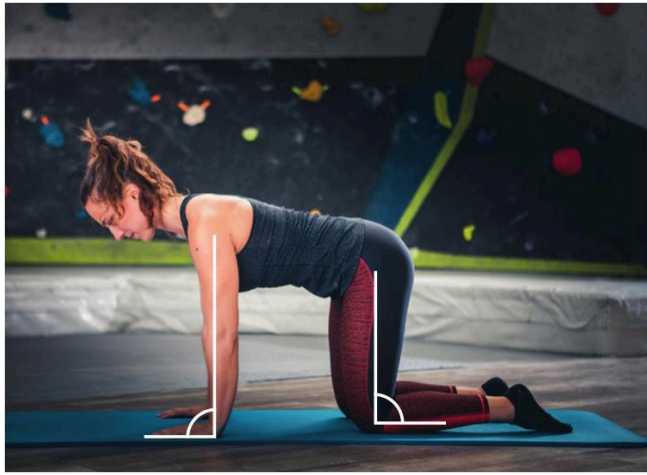
GLUTE BRIDGE

1



GLUTE BRIDGE

2



KNEELING SUPERMAN

1

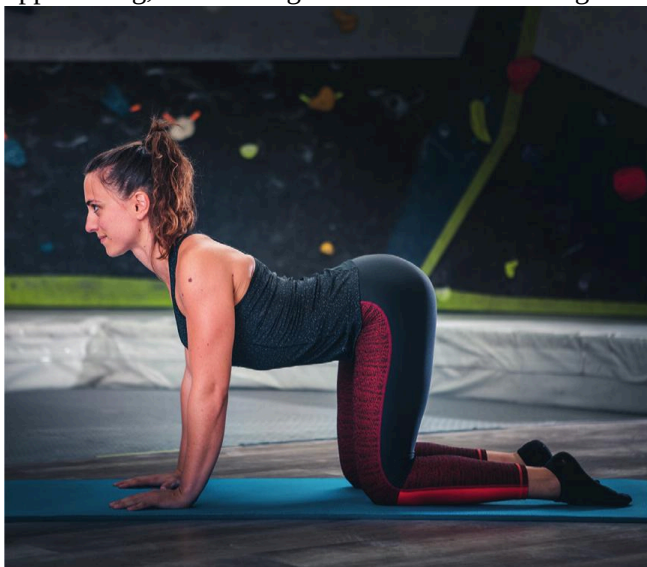


KNEELING SUPERMAN

2

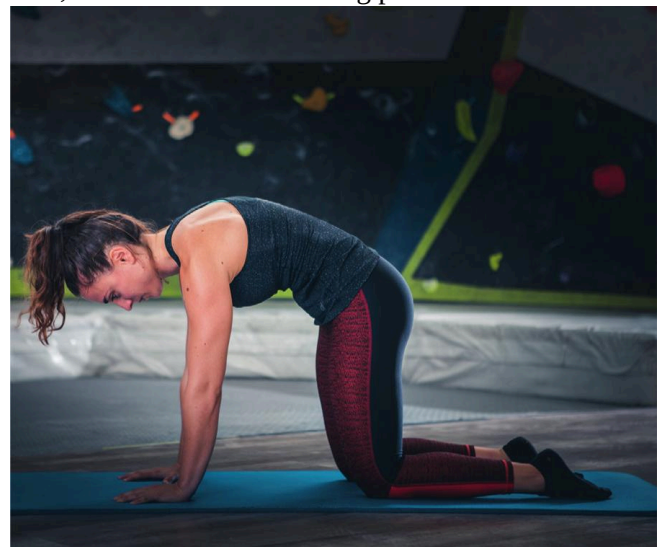
Glute bridge: a much better option than lower back extension. This exercise helps activate the hamstrings, glutes and paraspinal muscles in the lower back and encourages them to work in unison, which is how they're designed to function. Start by lying with your back on the floor, knees bent and feet hip-width apart, then raise your hips until your shoulders, hips and knees are aligned, stopping at this point.

Kneeling superman: this exercise requires good control of the shoulder girdle, abdominal corset, lower back muscles and hip extensors to keep the lower back in a neutral position. A good stability check is to balance a tennis ball on your lower back: if it falls off when you move, you know something isn't quite as stable as it should be. Begin on all fours with knees under hips and hands under shoulders. With a neutral spine and pelvis, extend one arm and the opposite leg, maintaining the natural curve through the lower back, then return to the starting position.



CAT-COW

1



CAT-COW

2

Cat-cow: cat-cow is a popular exercise that's used in all kinds of physical therapy and activities like yoga and Pilates to reprogramme spinal mobility. On all fours, tilt your pelvis forwards, hollowing out the lower back and exaggerating the natural curve of the spine, while retracting the scapulae and lifting the chin. Then do the opposite: bring the chin to the chest, push down through the floor to separate the scapulae, rounding out the upper back, and tilt your pelvis back.

Lower body exercises:

Barbell squat: the multi-joint exercise par excellence, cornerstone of lower body training and great for the rest of the body as well. You could write a whole thesis (or more) on how to do a squat, so I'll try not to get too carried away here. Before we begin, bear in mind that a squat will look and feel different for everyone. For example, the shape of your pelvis, length of your femur and your level of mobility will affect the width of your base (how far apart you have your feet), the angle of your feet (toes forwards or turned out), the depth of your squat (how low you can go),

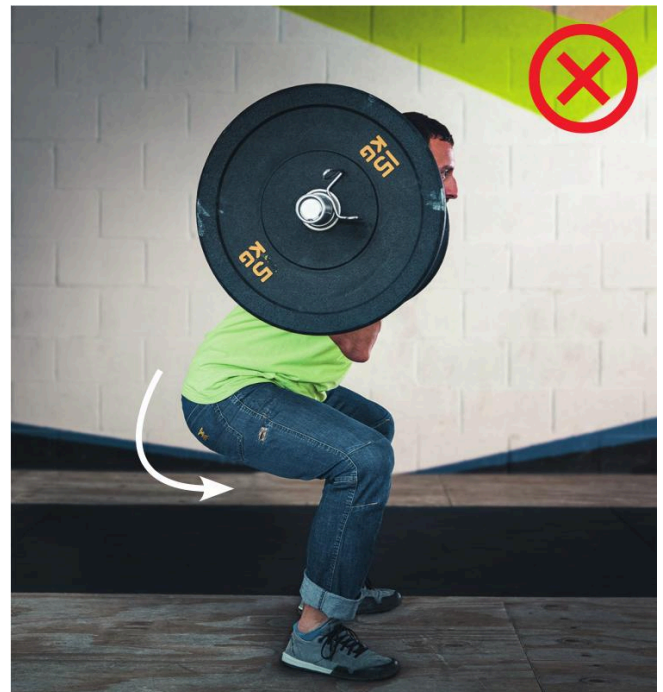
and so on. The most important thing is to be comfortable, learn the correct movement and then start training with an amount of added weight which is appropriate for you. Now, let's analyse the movement from the feet up:

To maintain stability, you need to create a solid base, so you should never, under any circumstances, lift your heels off the floor. The tibia should be 30 to 35 degrees off vertical. Unless you have any type of knee injury (in which case, you'll need a physiotherapist to advise on the ideal depth or degree of flexion), your knees should go over your toes when squatting. If not, your upper body will compensate by leaning too far forwards, which increases strain on the lower back by up to 1,000 per cent (Fry *et al.* 2003). McKean *et al.* (2010) analysed the biomechanics of the squat in trained participants and found that knees went over toes by approximately 7 centimetres in male participants and 9 centimetres in female participants. Another standard rule, this time for keeping the hips neutral, is that the knees should move directly in line with the toes, neither turning out nor in. Ensuring that the pelvis remains neutral will maintain the natural curve through the lumbar spine and prevent 'butt wink' (a posterior pelvic tilt caused by poor hip mobility when we squat lower than we ought to). This neutral position will ensure correct alignment of the intervertebral discs and lower the risk of injury.

If you don't have the mobility to squat past 90 degrees while keeping a neutral lower back and pelvis, Boyle (2010) suggests using a 2-centimetre heel lift. This reduces tension in the posterior chain and stops the hamstrings from pulling so much on the pelvis and causing butt wink. As for the upper body, the scapulae should be retracted and the barbell should sit on your trapezius, not your neck or your scapulae (Marchante, 2015). The elbows, depending on shoulder mobility, should point down or very slightly backwards. In summary: comfortable stance, knees forwards, hips back, chest up, squat down, stand up.



BARBELL SQUAT



BUTT WINK



BARBELL SQUAT

1



BARBELL SQUAT

2

Deadlift: this exercise really targets the posterior chain through both the upper and the lower body. In fact, some authors class it as leg training and others as back training (Delavier, 2004; Bompa and Cornacchia, 2002). Thanks to the incredible leg strength gained from this exercise, it's also recommended for climbers in the most recent edition of Eric Hörst's book *Training for Climbing*. In contrast to a squat, where you start high, go down (eccentric phase) and come up (concentric phase), in a deadlift you start low, go up (concentric phase) and come down (eccentric phase). To begin, stand with your feet hip-width apart (or however you're most comfortable), with toes pointing forwards. Your hands should be as close together as possible, without forcing the knees to turn in. Position the bar very close to your shins so that it touches when you bend down. Keeping your back perfectly straight, bend at the hips until you're almost parallel to the floor and then bend the knees to reach down and pick up the bar. With a firm grip on the bar, externally rotate your arms (elbows pointing back, biceps forwards), retract your scapulae and push out your chest. Lift the hips slightly to engage the hamstrings and push up through your legs. When you've got the weight above your knees, keep your back straight and hinge at the hips until you're standing up. To lower, keep the bar close to your body and start by hinging at the hips. When you've got the weight below your knees, bend your knees and place the bar back on the floor (Rippetoe, 2011; Nuckols, 2016).



DEADLIFT

1



DEADLIFT

2



DEADLIFT

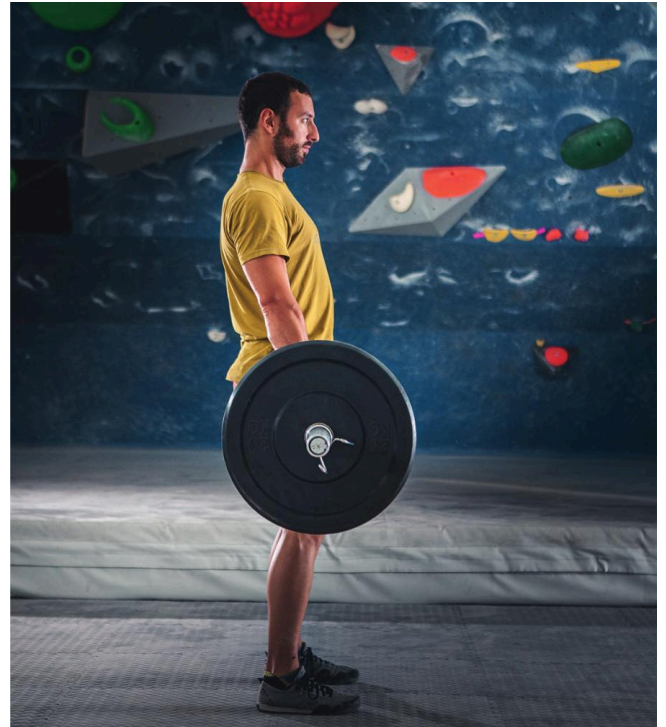
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ROMANIAN DEADLIFT

1



ROMANIAN DEADLIFT

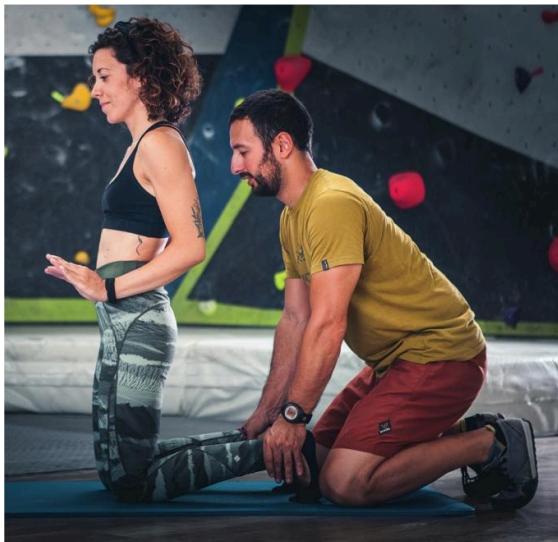
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ROMANIAN DEADLIFT

3

Romanian deadlift: unlike a regular deadlift, this lift starts from an upright position and you don't bend at the knee. Start at the top of a traditional deadlift (arms externally rotated, scapulae retracted, back straight) and slide the bar as far down your legs as you can without bending your knees. Both the tightness of your hamstrings and the straightness of your back will determine how low you can go. Instead of dipping your head, think about pushing your hips back as the bar glides down your legs. Hinge at the hips to return to standing. The closer the bar to your legs as you stand up, the more you'll activate the latissimus dorsi.



NORDIC CURL

1



NORDIC CURL

2



NORDIC CURL

3

Nordic curl: research shows that the Nordic curl is one of the best exercises for injury prevention. Van Dyk *et al.* (2019) performed a systematic review of the Nordic curl. They analysed 8,459 male and female athletes from different sports, aged 18 to 40, and found that athletes who did Nordic curls had a 51 per cent lower injury rate. As an exercise with an incredible eccentric load, it develops high levels of strength in the target muscles which, as we've said before, is the most efficient strategy for injury prevention. Begin by kneeling (ideally on some padding) with the feet firmly anchored to the floor. Keeping a straight line from the head to the knees (without bending at the hips), fall forwards at a slow, constant speed and engage your hamstrings to control the fall.



MODIFIED RAZOR CURL

1



MODIFIED RAZOR CURL

2

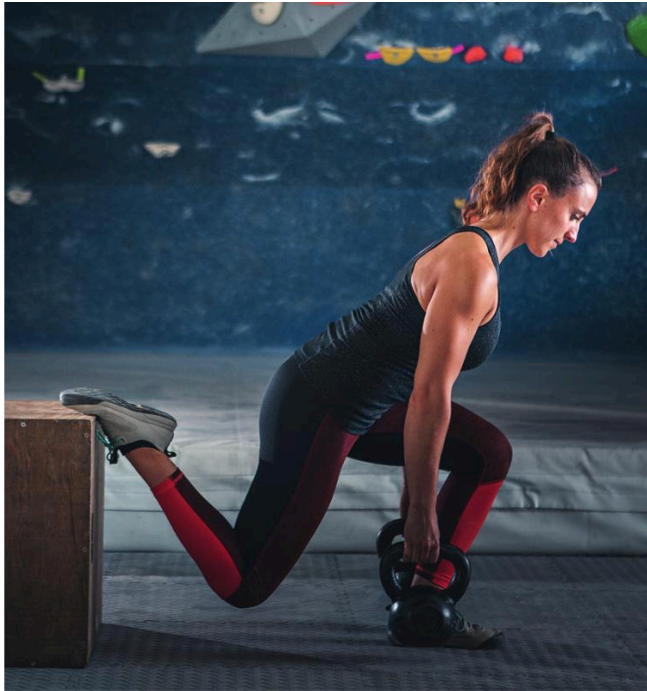


MODIFIED RAZOR CURL: HANDS ON CHEST



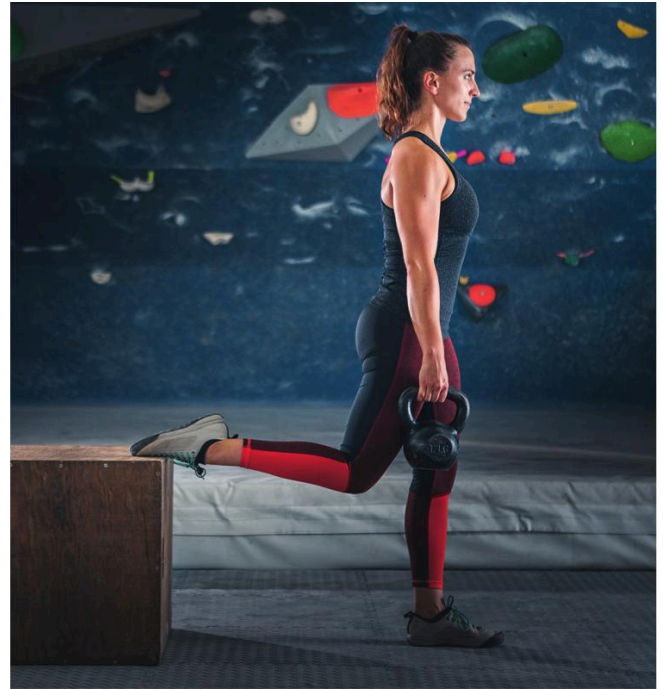
MODIFIED RAZOR CURL: HANDS ON HEAD

Modified razor curl: this variation of the razor curl is a combination of the Nordic curl and the Romanian deadlift. In fact, you could say it's a Romanian deadlift on your knees. Start as you would for the Nordic curl, with feet well anchored to the floor, keep your knees bent at 90 degrees and hips in line with the knees. Place your hands by your hips to reduce the load, or on your chest or head to increase the load (instead of adding weight, you're simply moving the 'mass' further away from the pivot point, which increases the load on your hamstrings as more mass has moved away from the pivot point). Lower your torso as much as you can, hinging at the hips and keeping the back completely flat, then come back up to the starting position. If this is too difficult, start by just lowering down, and as you get stronger you can work on coming back up.



BULGARIAN LUNGE

1



BULGARIAN LUNGE

2

Bulgarian lunge: this unilateral exercise really activates the glutes and the quads. From a standing position, support your back foot on a bench or step without hollowing out through the lower back. Hinge forwards slightly so the psoas doesn't compress the intervertebral discs in the lumbar spine. All your weight should be on your front leg – the aim of this exercise is to do a single-leg squat in a way that's biomechanically more similar to a regular squat. Lower your hips as much as possible, moving straight down not forwards (imagine you're going down in a lift). The front knee should drop to just level with the supporting foot. Come back up until the front leg is fully extended.

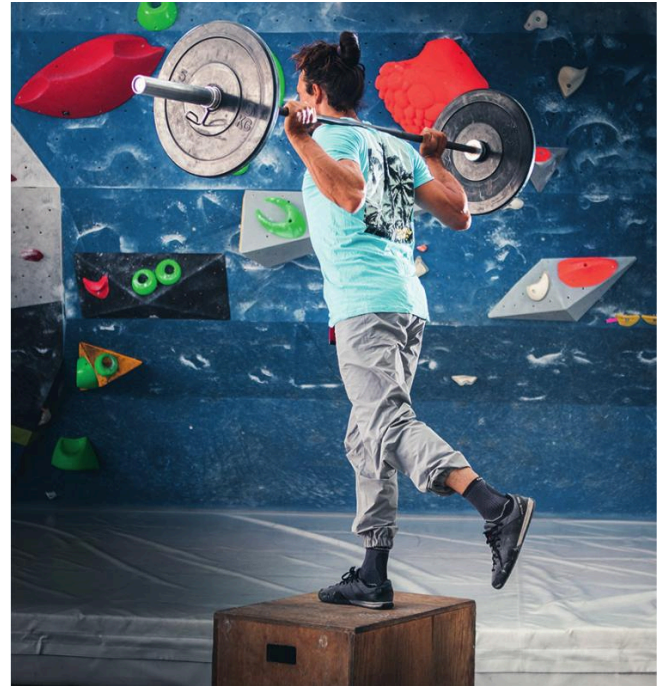
Step-up: a real test for the hip muscles, targeting both the gluteus maximus (the main muscle worked by this exercise) and the stabilising muscles. With a step in front of you, place one foot on the step. Step up to fully extend the working leg and tap the step lightly with the other foot. Depending on your balance, you could add extra weight with a barbell on your shoulders or by holding weights at chest height or on either side of your body. The higher the step, the more demanding it is on your glutes and the less demanding it is on your quads. This exercise has great transfer to climbing, especially for developing the strength required to weight a really high foothold.

Pistol squat: a variation or progression of the last two exercises. The pistol squat is a single-leg squat. To build up to a pistol squat, I'd strongly recommend starting with a bench behind you. Begin with a single-leg squat down to a seated position on the bench, lowering in a controlled, steady manner. The next step is to work on getting up from the bench. Gradually reduce the height of the bench until you don't need it any more and you can complete both the eccentric (downward) phase and the concentric (upward) phase of this exercise.



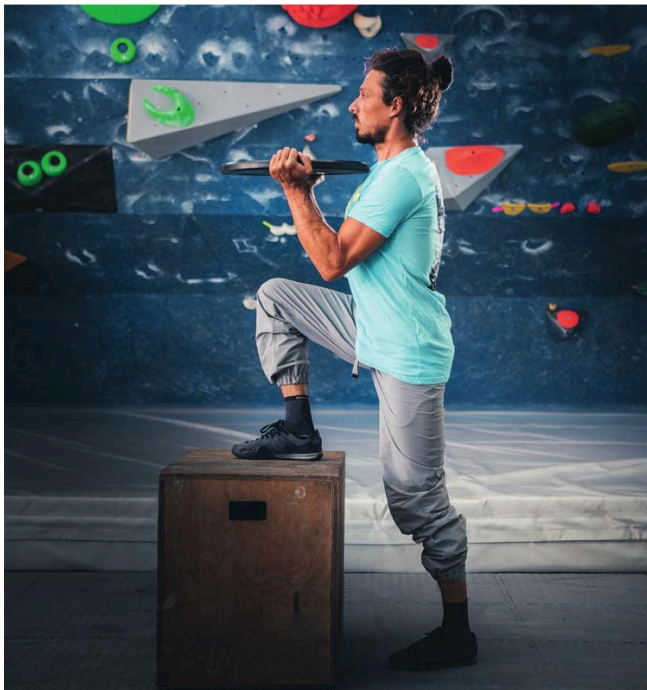
STEP-UP WITH BARBELL

1



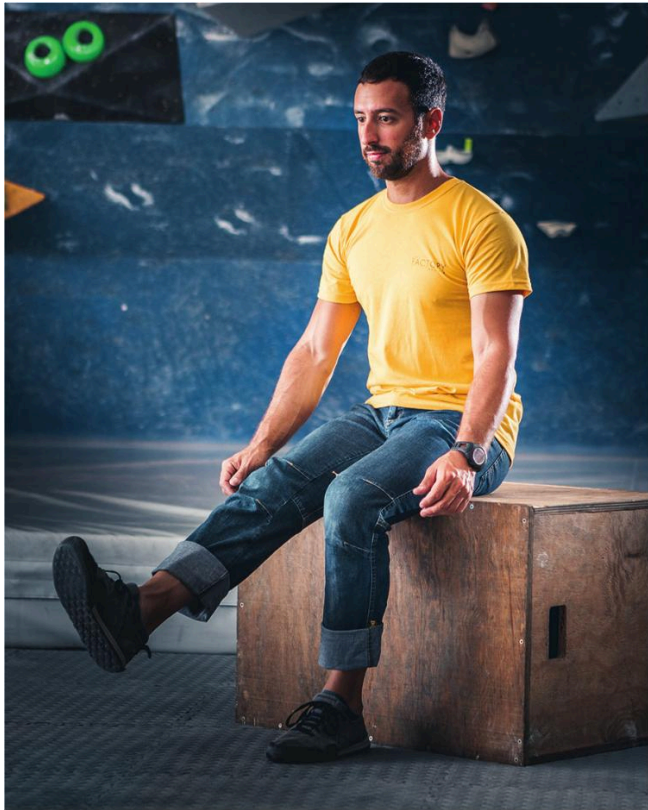
STEP-UP WITH BARBELL

2



STEP-UP WITH HANDHELD WEIGHT

Calf raises: stand with the balls of your feet on a step. Then alternate between dorsal flexion (dropping your heels as low as possible) and plantar flexion (raising your heels as high as possible). You can do one foot at a time or both at the same time. Heels together and toes apart will target the inner calf (the medial head), while toes together and heels apart will target the outer calf (the lateral head). Keeping your feet parallel will spread the load evenly across the calf. You can hold a plate, kettlebell or dumbbell in one hand and use the other hand for balance.



PISTOL SQUAT PROGRESSION

1



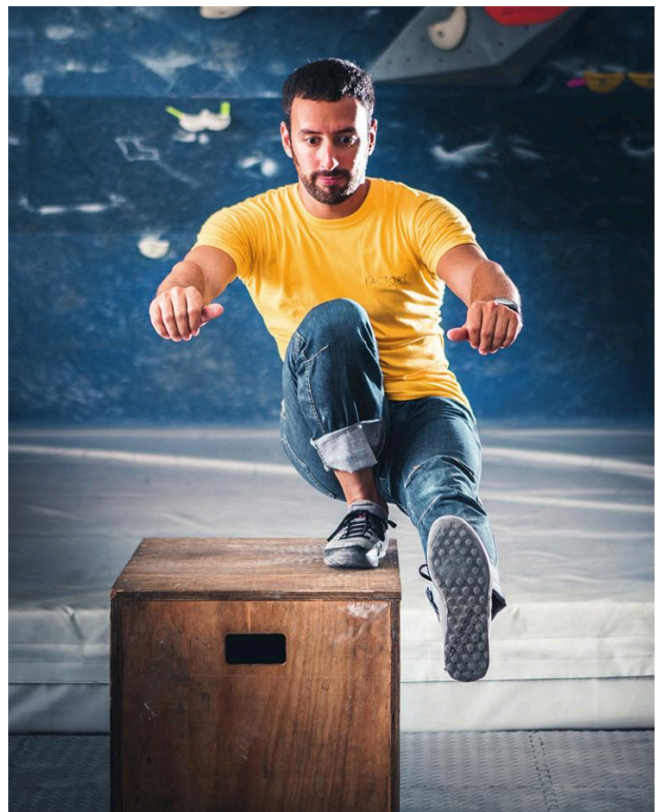
PISTOL SQUAT PROGRESSION

2



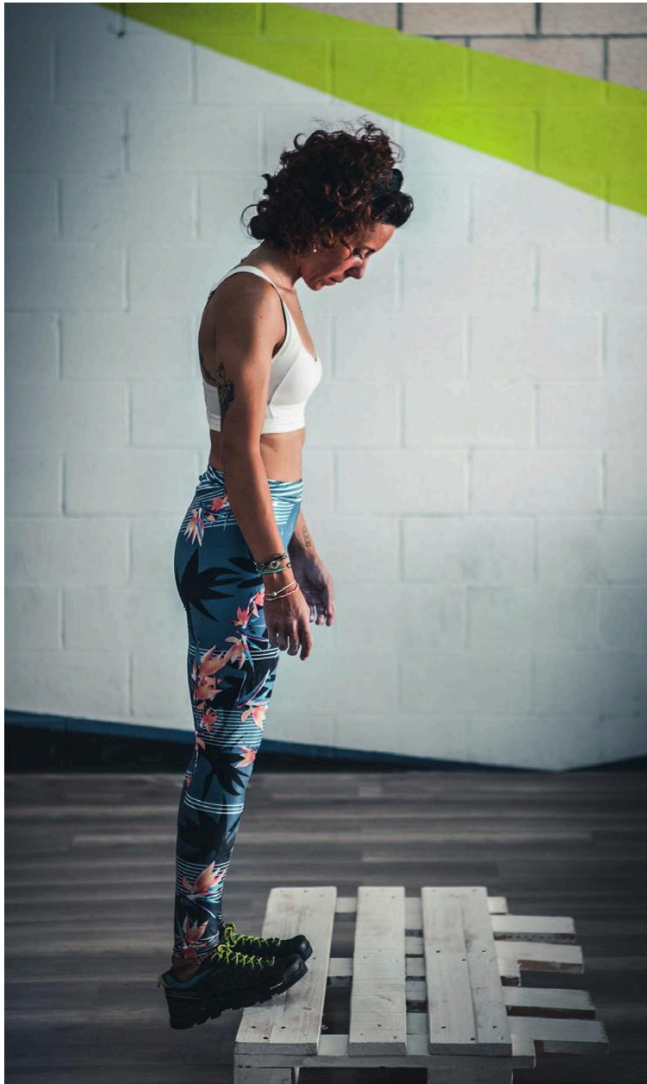
PISTOL SQUAT

1

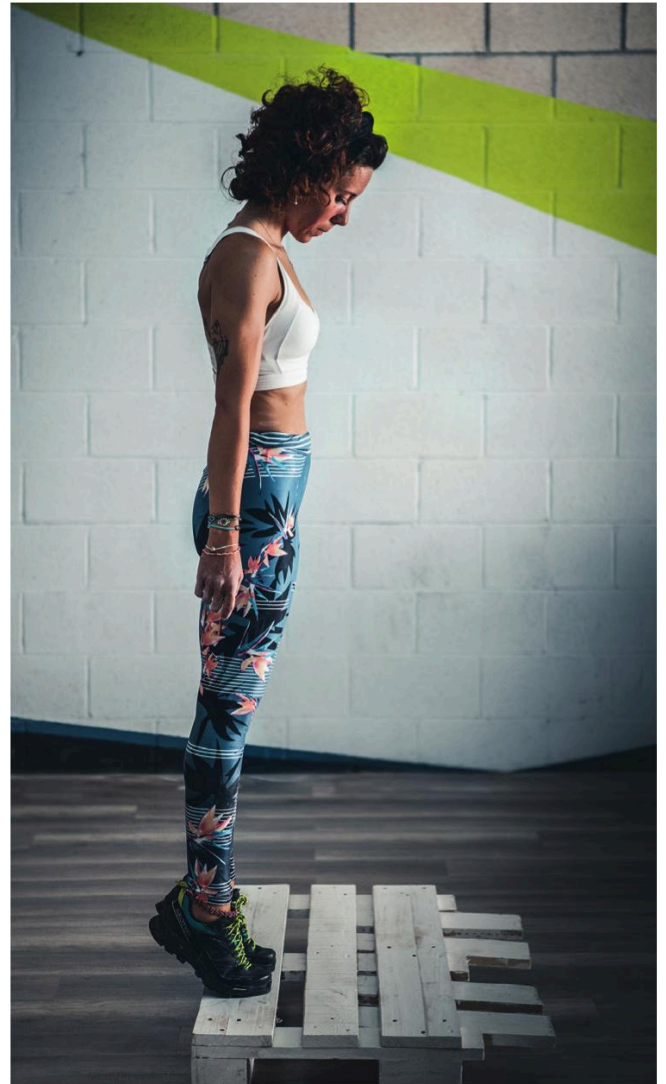


PISTOL SQUAT

2



CALF RAISES



CALF RAISES





SOLEUS

1



SOLEUS

2

Soleus: as seen in [chapter 4](#), the soleus is activated when the knees are bent. So, in a seated position, resting the balls of your feet on a step, place a weight (barbell, plate, free weight) on your knees and alternate between raising and dropping your heels.

STRENGTH TRAINING FOR INJURY PREVENTION

Injury is the biggest danger for any athlete, whether amateur or professional. A climbing injury might temporarily stop you from climbing, but it can also have very serious consequences, depending on where it is and how serious it is. It can even be life-threatening. A review conducted by Jones *et al.* (2018) found an injury rate of 2.71 ± 4.49 injuries per 1,000 hours of climbing. This may not seem that high, but in a study by Jones *et al.* (2008), 50 per cent of climbers had experienced at least one injury in the previous year. To begin, let's distinguish between three different types of injury:

Chronic or overuse injuries: Grønhaug (2018) and Jones *et al.* (2008) agree that this is the most common type of climbing injury and they found a significant correlation between probability of injury and climbing grade. That is to say, climbers operating at a higher grade, especially boulderers, have a higher rate of injury due to overuse/excessive load. This could be down to the amount of training they do and/or the intensity of the moves on harder problems. Schöffl *et al.* (2015) conducted an interesting study that assessed and compared which were the most common climbing injuries. They found that finger injuries were by far the most common (52 per cent), followed at quite a distance by shoulder injuries (17.2 per cent), hand injuries (13.1 per cent) and elbow injuries (9.1 per cent). Pulley injuries were the most common type of finger injury (15.4 per cent), mainly involving the C4, A4, A2 and A3 pulleys, and there was also a high percentage of capsulitis (9.5 per cent) and tenosynovitis (8.8 per cent).

Trauma: a fall, bump or blow to the body. The percentage of these injuries is significantly lower (10 per cent), although 28 per cent of trauma injuries are caused by moves that are too difficult or demanding (Jones *et al.*, 2008).

Postural changes caused by training: as with asymmetrical or unilateral sports, like tennis or javelin, where imbalances can occur between one side of the body and the other, certain postural 'deformities' can also arise in climbing. They mainly affect the spine and are referred to collectively as 'climber's back' (Förster *et al.*, 2009). As you'd expect, this adaptation/deformity is more pronounced in more advanced climbers. The upper back typically becomes rounded or hunched (kyphosis), causing the lower back to curve inwards (lordosis). As seen in [chapter 5](#), if our joints are misaligned in any way, they become less efficient and are more prone to wear, which in turn can lead to chronic injury.

Having reviewed the most common climbing injuries, let's now look at how to prevent them, focusing on injuries caused by overuse and postural changes. As for physical trauma, this type of injury is perhaps the most difficult to predict and prevent. However, with the figures that we've looked at, we can take solace in the fact that this doesn't happen very often.

The most common climbing injuries – finger injuries – are caused by overuse of the hands, yet you can hardly not use your hands when climbing! As such, the best advice is to gradually introduce and build up external load in your

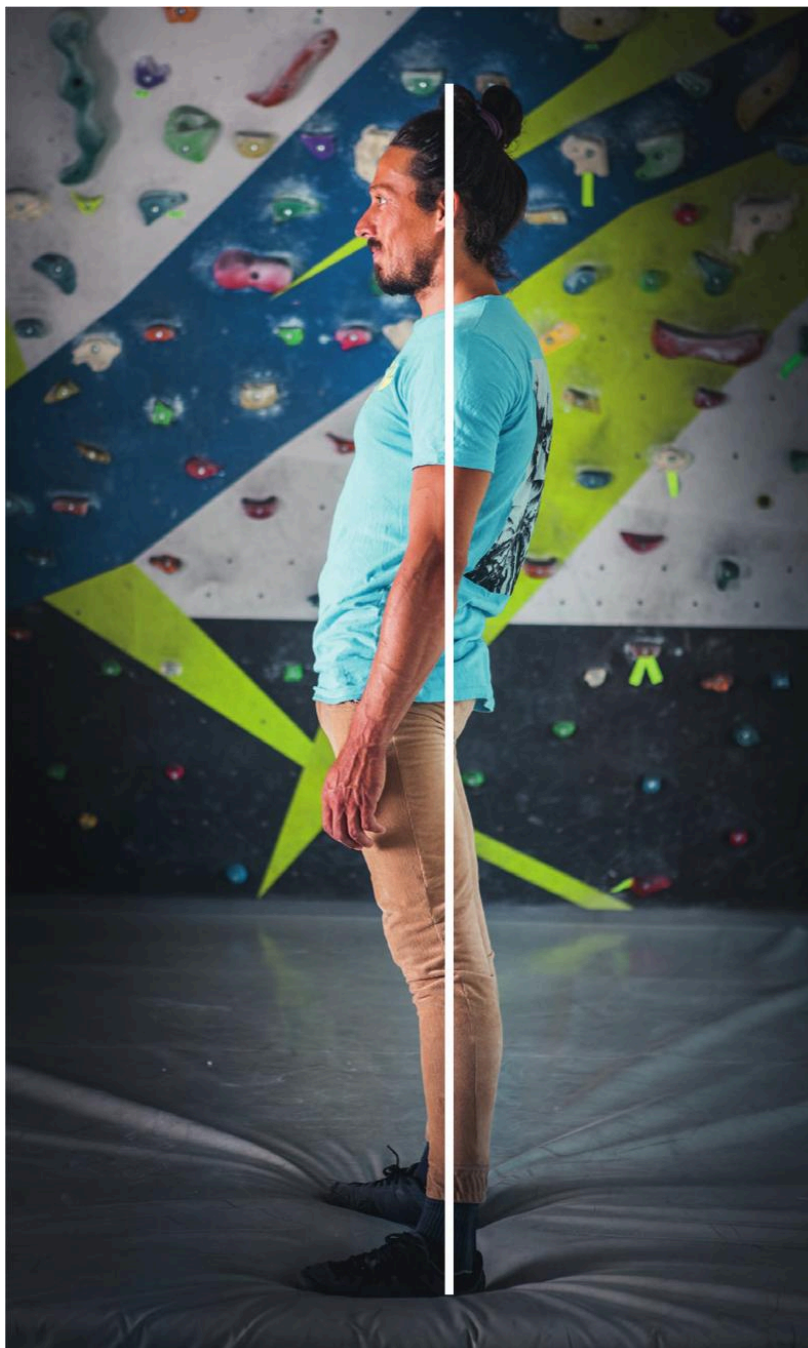
training and to respect recovery times and the impact of internal load, which is something we don't always do. Even then, the physical demands of training and climbing can be too much on our body and lead to injury. As such, we need to know how and when to recognise this overload, before it results in a functional injury. Think for moment about a car – a typical, ordinary car. You'd only take it to the garage if it was broken, or at the very most for its MOT. However, a Formula 1 car is on the ramps all the time, even when it's not broken, with a team of mechanics working to prevent any problems out on the track. If you want to perform like a Formula 1 car, you need to treat your body like a Formula 1 car: go to a physiotherapist and let them get to work. If you're injured, they'll map out your recovery, steer you away from disaster and guide you back into training. As every injury is a world unto itself and needs to be assessed and treated by a healthcare professional, rehabilitation is not an area covered by this book. As such, we're going to focus solely on prevention, which in any case is always better than cure. However, having covered what causes climbing injuries, we'll also look at how and why they occur. This will make it easier to identify any process or exercise that may end up causing an injury.

Tendons: as seen in [chapter 4](#), the main job of our tendons is to transmit mechanical force. When they are repeatedly loaded, tendons experience what is known as fatigue damage. With sufficient recovery time, the body can compensate for and repair this damage, returning the tendon to a state of homeostasis. However, with insufficient recovery time and continued loading, or if there's a sudden increase in compressive or tensile load that the tendon isn't able to absorb, the cells in the tendon respond by drastically increasing in number and in water content. This reduces the load by increasing the diameter of the tendon. It's important to stress that this increase in diameter is an attempt to reduce load on the tendon, as per Young's modulus strategy, and not an inflammatory response (which is why ice and anti-inflammatories are of little use against tendonitis). If the load doesn't stop and this strategy fails, the cells will start producing extra collagen and proteoglycans. This destructures and disorganises the cell fibres, causing a pathological thickening of the tendon. If this process continues, it can cause almost irreversible damage with the onset of cell death and the appearance of new blood vessels, discontinuous fibres, and so on, leading, in 97 per cent of cases, to an inevitable rupture of the tendon (Cook *et al.*, 2017; Wiesinger *et al.*, 2015). This could be either a creep rupture if caused by progressive degeneration, or a crack rupture if caused by the accumulation of micro-cracks, leading to a full rupture or tear (Ker, 2007). Interestingly, a study by Earp *et al.* (2016) found that strength exercises at a slower tempo, as opposed to at maximum concentric speed, resulted in increased tendon compression at the start and end of the movement, causing increased time under tension and greater damage and degradation. This is another good reason for doing strength exercises at maximum speed and for questioning those super-slow pull-ups favoured by some coaches.

Pulleys: pulleys are fibrous sheaths that closely tether tendon to bone. Imagine a fishing rod: the rod is the finger bone, the line is the tendon and the rings that guide the line along the rod are the pulleys (see image of the pulley system in [chapter 4](#)). Crimping puts a lot of stress on the pulleys (mainly the A2, which plays a key role in finger flexion) and can cause either a full or partial tear, or gradual stretching in the case of repetitive stress (Font *et al.*, 2005). After studying the biomechanics of climbers' pulleys, Schweizer (2008) confirmed that crimping can result in tenosynovitis and a full or partial rupture of the A2 and A4 pulleys, especially in the absence of a proper warm-up of at least 100 moves. In turn, Schreiber *et al.* (2015) studied pulley adaptations in a group of F7b+ to F9a climbers compared to a group of non-climbers. In climbers, the A2 and A4 pulleys were 62 to 76 per cent thicker and the tendons were up to 21 per cent thicker. These adaptations occurred after 15 years of climbing and training for climbing, so don't expect changes overnight!

Exercises to curb the excesses caused by climbing, including the classic 'climber's back' and any form of joint instability, should be a common feature of any training programme. This is known as antagonist or compensatory training.

A 'climber's back' is synonymous with sway-back posture (upper and lower crossed syndrome) and with excessive tension in the AM and AP [muscle chains](#). Regardless of what it's called, this condition involves an excess of muscle tone (hypertonia) in certain muscles (that need relaxing), which pulls segments of bone out of place and deforms our posture. This is linked to a lack of muscle tone (hypotonia) or inhibition in the antagonist muscles, which can't counter the excessive force of the agonist muscles and therefore can't stabilise the joint.



SWAY-BACK POSTURE

The table on the next page shows hypertonic muscles (to relax) and hypotonic or inhibited muscles (to strengthen) (adapted from PowerExplosive, 2017). Remember that the agonist/antagonist pairing isn't always as simple as front/back, biceps (flexor)/triceps (extensor). If you're unsure about this, take another look at [chapter 4](#). It explains the key muscles and joints in the human body and will help you steer clear of bro science and popular myths. For example, the common misconception that the antagonist of the latissimus dorsi (a key muscle in climbing) is the pectoralis major: both are in fact internal rotator muscles. So, if to balance out the latissimus dorsi you go to town on the pectoralis major, you'll only be making your sway-back posture worse and destabilising the shoulder joint.

Hypertonic muscles	Inhibited muscles [†]
Latissimus dorsi	Middle and lower trapezius
Pectoralis major	Rhomboids
Anterior deltoid	Serratus anterior
Anterior trapezius	Longus colli

Hypertonic muscles	Inhibited muscles [†]
Sternocleidomastoid	Transversus abdominis
Quadratus lumborum	Multifidus
Erector spinae muscles	Gluteus maximus
Rectus abdominis	Infraspinatus
Iliopsoas	Teres minor
Rectus femoris	Semitendinosus
	Semimembranosus

For the best results from antagonist training (and to not cause a fight between one muscle and another), you need to relax or inhibit the hypertonic muscles (see [chapter 3](#) on mobility) before working on the hypotonic muscles. The exercises in the previous section (plank, glute bridge, kneeling superman, face pull, Romanian deadlift and modified razor curl) are essential for helping to correct ‘climber’s back’.

Finally, and although it’s not overly common, I feel it’s important to dedicate a few lines to the prevention of medial and lateral epicondylitis.

Medial epicondylitis or golfer’s elbow: this is a proximal tendinopathy caused by overloading the flexor muscles of the wrist (flexor carpi ulnaris, flexor carpi radialis and palmaris longus), although it can also affect the biceps brachii. It can be treated by releasing tension through self-massage and physiotherapy (Nogueiras, 2016). Golfer’s elbow can lead to excessive strain on the extensor muscles, which have to fight against the strong pull of the flexor muscles to keep the wrist stable, resulting in reactive lateral epicondylitis.

Lateral epicondylitis or tennis elbow: unlike golfer’s elbow, tennis elbow is a proximal tendinopathy caused by excessive tension in the extensor muscles of the wrist, mainly the extensor carpi radialis brevis. In terms of treatment, strengthening exercises have proved more effective than corticosteroid injections (Araya Quintanilla and Moyano Galvez, 2015). Kenas *et al.* (2015) propose doing at least two wrist extension exercises (such as wrist curls with added weight and eccentric pronation) three times a week, spending four to six seconds on the eccentric phase of the exercise (at a constant speed) and progressing, pain dependent, to loads of 70 to 80 per cent 1RM.

CARDIOVASCULAR ENDURANCE TRAINING FOR CLIMBING. BASE ENDURANCE, HIIT, BODY COMPOSITION AND SIT

Do you need to run/swim/cycle (or do any other form of cardio) to get better at climbing? Does cardio have any real transfer to climbing? In truth, unless you’re trekking for hours to get to the crag, aerobic endurance isn’t really a limiting factor in any type of climbing. So, you shouldn’t need a good level of cardiovascular endurance to get better at climbing. However, in reality, it isn’t quite that simple. Endurance training can have a positive impact on two aspects of climbing performance:

Base endurance: this can be defined as the ability to perform any type of activity, in any given sport, involving multiple muscle groups and systems (central nervous system, cardiovascular, respiratory, and so on) over a prolonged period of time, with an optimal effect on specific performance. It’s a generic type of endurance needed in any sport that isn’t specifically an endurance sport (such as trail running, or cross-country skiing). Cardiovascular endurance is measured using VO₂ max, or maximal oxygen consumption. By definition, this is the maximum amount of oxygen we can get into our lungs, transport round the body and use to produce energy. Although they only had a small number of participants, a study by Billat *et al.* (1995) found that F7b climbers had a VO₂ max of 54.8 ± 5 ml/kg/min, which we could use as a benchmark for optimal base endurance. That is, if you do a cardiopulmonary exercise test (effort test) and your VO₂ max is lower than this, you might want to work on your base endurance. However, exceeding this benchmark isn’t going to have a big effect on climbing performance, as even on the hardest routes, you’re only using a fairly small fraction of your maximal aerobic capacity (Sheel, 2004).

How can you improve your VO₂ max? On the one hand, VO₂ max is largely determined by genetics (Wilmore and Costill, 2007); that said, the VO₂ max recorded by Billat *et al.* (1995) is not completely unattainable for the average person. On the other hand, the oh-so-popular HIIT (high-intensity interval training) is the most efficient way to increase VO₂ max for both elite athletes and normal people, including people with injuries or illness, young people or adults (Mallol *et al.*, 2019; Batacan *et al.*, 2017; García-Hermoso *et al.*, 2016). Although running is probably the simplest way to get the best results, if this isn’t an option because of your knees, level of ability or any other reason, pick your activity (rowing, cycling) and be prepared to sweat. According to a meta-analysis performed by Wen *et al.* (2019), the best results are achieved with intervals of over two minutes, total work time (sum of the

intervals) of over 15 minutes and training blocks of four to 12 weeks. With HIIT, you need to be working at maximum intensity, aiming for above 90 per cent of maximum heart rate, power, and so on, for the entire workout. Maintaining this level of intensity for a full 15 minutes isn't easy and will require some training. To build up to it, try playing with the rests between intervals: progressing from a 1:3 work:rest ratio to a 2:1 ratio.

The following example is for running training for beginners (in HIIT, not climbing):

7 sets × 2 minutes of running at 90–95 per cent of max heart rate × 6 minutes of active rest (easy jogging) at 60–70 per cent of max heart rate

And for advanced training:

10 sets × 2 minutes of running at 90–95 per cent of max heart rate × 1 minute of active rest at 60–70 per cent of max heart rate

Body composition: another reason that people take up running is to maintain a low body fat percentage, to burn calories or to lose weight. Before explaining why cardio is a hugely inefficient way to lose weight, let's look at what constitutes ideal body composition for climbers. This will help us to set achievable, sensible targets for improving body composition. Fortunately, Couceiro (2010) covered this exact topic as part of his PhD thesis. He established the following parameters for three different groups:[±]

- a) *Elite*: male, onsight F7b+/8a, redpoint F8a/8c and boulder Font 7b/8a+, weight 65.8 ± 7.5 kg, body fat 8.8 ± 2.6 per cent.
- b) *Advanced*: male, onsight F6c+/7a+, redpoint F7a+/7c+ and boulder Font 6b/7a+, weight 67.2 ± 5.5 kg, body fat 9.9 ± 4 per cent.
- c) *Female*: onsight F6a/7a+, redpoint F6b/7b+ and boulder Font 6b/7a, weight 51.6 ± 4.5 kg, body fat 19.26 ± 3.5 per cent.

Without wishing to oversimplify what is a complex and sensitive topic in climbing, these parameters could allow you to measure your body fat percentage and sensibly establish how much fat you might want to lose to climb at a desired grade.

Let's return to how inefficient cardio is for weight loss and take an objective look at the numbers: meet Paco, a climber who took part in the study by Billat *et al.* (1995). Paco's VO2 max is 54.8 ml/kg/min, he weighs 71 kilograms and his max heart rate is 205bpm (which were the average measurements in this study). You could say that his absolute VO2 max is $54.8 \times 71 = 3.9$ L/min. VO2 max is directly proportional to maximum heart rate, meaning that if Paco runs at maximum heart rate, he'll also be running at VO2 maximum and if he runs at 80 per cent of his maximum heart rate, he'll be running at 80 per cent of his VO2 max. In turn, we know that one litre of oxygen yields 5kcal of energy, one gram of carbohydrate yields 4kcal and one gram of fat yields 9kcal. Imagine that Paco runs for an hour and, as he wants to burn lots of calories, he runs at a fast pace for the full hour: 175bpm and 85 per cent of his max heart rate. Let's see what the numbers say: if Paco runs at 85 per cent of 3.9 L/min, he's consuming 3.31 L/min and expending $3.31 \times 5 = 16.6$ kcal/min, making for a total of 16.6×60 minutes = 996 kcal. Even if Paco was burning 100 per cent fat (which is completely unviable as such high intensities produce large amounts of H⁺, which inhibits lipolysis), he would have burnt $996/9 = 111$ grams. One hour of running at 175bpm, with the fatigue this would entail, for just 111 grams. And what's more, this is based on very high levels of VO2 max, intensity and activity time. As you can see, cardio is not a very efficient way to burn fat.

So, what is the most efficient way to lose weight and lower body fat percentage? The key is to control your insulin levels and this means being very aware of what you eat. As nutrition is a vast and complex topic, which is beyond the limits of this book, I recommend reading *The Obesity Code* by Dr Jason Fung. This book explains why all diets work and why none of them provide a long-term solution. It also explains how to keep insulin levels in check by managing your intake of macronutrients and through intermittent fasting. That said, physical activity such as interval training does undoubtedly play a part in burning calories and reducing body fat percentage. SIT (sprint interval training) involves shorter intervals than HIIT (5, 15 and 30 seconds), and it uses a work-to-rest ratio of 1:8 (40, 120 and 240 seconds of rest, respectively), adding up to a total work time of two to three minutes going all-out, at maximum intensity (McKie *et al.*, 2018). There's no evidence to suggest one length of interval is more efficient than the others for burning fat. However, 30-second intervals create a higher increase in VO2 max (nearly double) than 5-second or 15-second intervals (McKie *et al.*, 2018), which is good to know if you want to use SIT to improve body composition and base endurance at the same time.

MOBILITY TRAINING FOR CLIMBING

In truth, there's not much to say about mobility training that's not already been said in previous chapters. However, I would like to distinguish between two different purposes of mobility training. One is to correct postural imbalances that can lead to joint misalignment, with the aim of preventing excessive wear and compensation elsewhere in the body, as per the biotensegrity principle. The other is to increase range of motion to optimise movement and technical ability (for example, the ability to get your foot above your hip to use a high foothold).

First, let's recall what mobility training is really about: reassuring the brain and ensuring that any new range of motion is functional (force can be applied within it). This highlights the need to include mobility in our wall-based training sessions, at least if the aim is to increase useful range of motion for climbing. However, if the aim is to correct postural imbalances or to reduce excessive tension in certain muscle chains, it might be best to save mobility training for days when you're not at the wall.

Mobility training to increase useful range of motion for climbing: as seen in previous chapters, the best and most logical approach is to include mobility training as part of your warm-up. If you can get the brain to expand your ROM as you warm up, and you then use this range of motion during your session, it will become part of how you naturally move on the wall. This will reassure the brain that you can move safely within this new ROM and, by repeating this experience of expanding and then using your newly acquired ROM, you can gradually convince the brain to keep this ROM open to you all the time. Remember: active stretching, foam rolling, massage balls, local vibration and flossing will improve ROM without triggering the autogenic inhibition reflex and inhibiting contractile capacity.

Mobility training to correct postural imbalances: physiotherapy here is key. On the one hand, your physiotherapist will use the most suitable techniques to relax muscle tissue and help reorganise bundles of muscle fibres, fascia cells, and so on, to promote healthy movement and function. Think back to the analogy of the Formula 1 car and the expert mechanics. On the other hand, your physiotherapist may also do a GDS muscle chain assessment, indicating which of your chains are causal or reactive and which muscles are being overworked. After finding out which areas you need to focus on, it's time to get to work.

First, you need to create a neurophysiological environment that facilitates mobility training and reduces the chance of triggering pain signals in the brain. Begin with some deep breaths to boost muscle oxygenation and to reduce high alertness and activation of the sympathetic nervous system. Be aware of room temperature (colder temperatures cause muscle tension/spasm and reduced blood flow due to vasoconstriction; hotter temperatures increase blood flow and reduce muscle tension), lighting (lighting has a significant effect on serotonin, which regulates our sleep/wake cycle), noise, and so on. Once you've got the right environment, begin with some exercises to reduce descending inhibition so that less pain information reaches the brain and you can increase your stretch tolerance. Use a foam roller or massage ball in areas with the most tension, aiming for 8 to 10 passes per area. After this, the central nervous system will be more receptive and better prepared for stretching. Now you're ready for some stretches to safely inhibit the target muscles. Let's look at a technique that gives a greater increase in ROM than passive stretching (Hindle *et al.*, 2012): proprioceptive neuromuscular facilitation (PNF). PNF covers two different methods: the CR method (contract/relax) and the CRAC method (contract/relax/antagonist/contract). In the CR method, the target muscle is stretched for a few seconds until you feel a light stretch. Once you start feeling the stretch, you isometrically contract the muscle and hold for 10 to 15 seconds. You then relax the muscle and stretch for a further 15 to 20 seconds. The CRAC method is an extension of the CR method: after isometrically contracting the stretched muscle (the agonist), instead of relaxing the muscle and stretching again, you contract the antagonist muscle for 6 to 10 seconds, then relax and go back to stretching the agonist muscle. For example, if you're stretching the hamstrings, first contract these muscles, then the quads. Depending on individual ability and tolerance, the study authors recommend two to four cycles per agonist muscle.

* To calculate the weight lifted in the Brzycki formula, use body weight only for pull-ups. If the formula is used to calculate RM for any other exercise, use weight lifted only.

† NB: This table shows hypertonic and inhibited muscles, implying no agonist/antagonist relationship between these pairings.

‡ Editor's note: it should be noted this study appears to have a questionable approach to gender equality, by only assigning 'elite' or 'advanced' designations to men, and combining all women into one category with a lower grade range.